

Yuexuan Xu, MPP, PhD

(Updated July 1st, 2026)

The Gertrude H. Sergievsky Center
Taub Institute for Research on Alzheimer's Disease and the Aging Brain
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RESEARCH INTERESTS

My research focuses on elucidating the genetic and molecular mechanisms underlying Alzheimer's disease, with an emphasis on developing genomic and multi-omic risk prediction models across diverse populations. I integrate large-scale genetic and genomic epidemiology studies with multi-omic data—including transcriptomic, metabolomic, proteomic, and epigenomic profiles—and biomarker measures such as plasma and CSF analytes (e.g., P-tau217) to improve early disease detection and risk stratification. My work also includes examining gene–environment interactions and applying advanced statistical and machine learning approaches to genomic and multi-omic prediction of longitudinal cognitive and disease trajectories, with the broader goal of informing precision medicine strategies for neurodegenerative diseases.

EDUCATION

- | | |
|-------------|---|
| 2019 - 2023 | Doctor of Philosophy in Population Health Sciences (Ph.D.)
Concentration: Statistics, Quantitative Genetics, and Genetic Epidemiology
University of Wisconsin-Madison (Madison, WI, USA)
Dissertation: <i>“Identification of Life-Course Gene-Environment Interactions in Cognitive Decline During Preclinical Alzheimer’s”</i>
Advisor: Dr. Corinne Engelman |
| 2016 - 2018 | Master of Public Policy (M.P.P.)
Concentration: Social Science Genomics
University of Southern California (Los Angeles, CA, USA) |
| 2011 - 2015 | Bachelor’s Degree (Management)
Shandong University (Jinan, Shandong, China) |

ACADEMIC POSITIONS

- | | |
|----------------|---|
| 2024 - Present | Postdoctoral Research Scientist (Mentor: Dr. Richard Mayeux)
<i>The Gertrude H. Sergievsky Center</i> |
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Taub Institute for Research on Alzheimer's Disease and the Aging Brain
Columbia University (New York, NY, USA)

- 2024 - Present **Honorary Associate**
Department of Population Health Sciences
University of Wisconsin-Madison (Madison, WI, USA)
- 2020 - 2023 **Research Assistant (Mentor: Dr. Corinne Engelman)**
Department of Population Health Sciences
University of Wisconsin-Madison (Madison, WI, USA)
- 2019 - 2020 **Research Assistant (Mentor: Dr. Jason Fletcher)**
Center for Demography of Health and Aging
University of Wisconsin-Madison (Madison, WI, USA)
- 2018 - 2019 **Research Specialist (PI: Dr. Titus Galama)**
Center for Economic and Social Research
University of Southern California (Los Angeles, CA, USA)
- 2018 - 2018 **Teaching Assistant (Instructor: Dr. John Romley)**
Sol Price School of Public Policy
University of Southern California (Los Angeles, CA, USA)

PUBLICATIONS

Peer-Reviewed Articles

1. **Xu Y**, Vasiljevic E, Deming YK, Jonaitis EM, Kosciuk RL, Van Hulle CA, Lu Q, Carboni M, Kollmorgen G, Wild N, Carlsson CM, Johnson SC, Zetterberg H, Blennow K, Engelman CD. Effect of Pathway-Specific Polygenic Risk Scores for Alzheimer's Disease (AD) on Rate of Change in Cognitive Function and AD-Related Biomarkers Among Asymptomatic Individuals. *J Alzheimers Dis.* 2023;94(4):1587-1605. doi: 10.3233/JAD-230097. PMID: 37482996; PMCID: PMC10468904.
2. **Xu Y**, Sun Z, Jonaitis E, Deming Y, Lu Q, Johnson SC, Engelman CD. Apolipoprotein E moderates the association between non-APOE polygenic risk score for Alzheimer's disease and aging on preclinical cognitive function. *Alzheimers Dement.* 2024 Feb;20(2):1063-1075. doi: 10.1002/alz.13515. Epub 2023 Oct 19. PubMed PMID: 37858606; PubMed Central PMCID: PMC10916952.
3. **Xu Y**, Qiao M, Gunasekaran TI, Gu Y, Reyes-Dumeyer D, Piriz A, Sanchez D, Soriano B, Franco Y, Coronado ZD, Recio P, Mejia DR, Medrano M, Lantigua RA, Honig L, Manly JJ, Brickman AM, Vardarajan BN, Mayeux R. Evaluating polygenic risk score prediction performance for Alzheimer's disease in a population-based Hispanic cohort using single- and multi-ancestry models. *Lancet Reg Health Am.* 2025 Jul 25;49:101198. doi: 10.1016/j.lana.2025.101198. PMID: 40747112; PMCID: PMC12312067.

4. **Xu Y**, Qiao M, Gunasekaran TI, Gu Y, Reyes-Dumeyer D, Piriz A, Sanchez D, Soriano B, Franco Y, Coronado ZD, Recio P, Mejia DR, Medrano M, Lantigua RA, Honig L, Wilson R, Manly JJ, Brickman AM, Engelman CD, Johnson S, Asthana S, Vardarajan B, Mayeux R. Polygenic scores of Core 1 Alzheimer's disease biomarkers predict early cognitive and pathological change. *Alzheimers Dement.* 2025 Nov;21(11):e70937. doi: 10.1002/alz.70937. PMID: 41307156; PMCID: PMC12658386.
5. **Xu Y**, Sun Z, Jonaitis E, Deming Y, Lu Q, Johnson SC, Engelman CD. Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals. *medRxiv* [Preprint]. 2024 May 27:2024.05.26.24307953. doi: 10.1101/2024.05.26.24307953. PMID: 38853902; PMCID: PMC11160812.
6. **Xu, Y.**, Gunasekaran, T. I., Gu, Y., Reyes-Dumeyer, D., Piriz, A., Sanchez, D., ... & Mayeux, R. (2026). Plasma phosphorylated tau 217 concentrations, APOE genotype, and timing of cognitive impairment in individuals across diverse racial and ethnic groups: A pooled analysis of prospective cohort studies. *The Lancet Neurology*. In press.

SEMINAR (INVITED) TALKS

- 2023 *“Beyond the Surface: Reassessing Genes, Environment, and their Interaction in the Preclinical Stage of Alzheimer’s.”*
- Department of Psychiatry and Behavioral Sciences, UCSF
 Department of Population Health Sciences and Epidemiology, Stanford
 Department of Genetics and Genomic Sciences, Icahn School of Medicine

CONFERENCE PRESENTATIONS

Peer-Reviewed Abstracts

- 2026 *“Pathology-Informed Polygenic Scores Identify Distinct Genetic Contributors to Alzheimer’s and Non-Alzheimer’s Dementia Across Ancestries.”* Accepted for poster presentation at the Clinical Trials on Alzheimer's Disease Conference 2026, Boston, MA, USA.
- 2026 *“Plasma P-tau217 and APOE Genotype Can Predict Onset of Cognitive Impairment Before the Diagnosis of Alzheimer’s Disease.”* Accepted for oral presentation at the Clinical Trials on Alzheimer's Disease Conference 2026, Boston, MA, USA.
- 2026 *“Pathology-Informed Polygenic Scores Identify Distinct Genetic Contributors to Alzheimer’s and Non-Alzheimer’s Dementia Across Ancestries.”* Accepted for poster presentation at the Alzheimer’s Association International Conference 2026, London, UK.
- 2026 *“Plasma P-tau217 and APOE Genotype Can Predict Onset of Cognitive Impairment Before the Diagnosis of Alzheimer’s Disease.”* Accepted for oral

presentation at the AD/PD Conference 2026, Copenhagen, Denmark.

- 2025 *“Polygenic scores of Core 1 Alzheimer's disease biomarkers predict early cognitive and pathological change.”* Accepted for poster presentation at the Alzheimer's Association International Conference 2025, Toronto, Canada.
- 2025 *“Polygenic scores of Core 1 Alzheimer's disease biomarkers predict early cognitive and pathological change.”* Accepted for poster presentation at the AD/PD Conference 2025, Vienna, Austria.
- 2024 *“Evaluating polygenic risk score prediction performance for Alzheimer's disease in a population-based Hispanic cohort using single- and multi-ancestry models.”* Accepted for poster presentation at the American Society for Human Genetics Annual Conference 2024, Denver, CO, USA.
- 2024 *“Sex Differences in the Risk of Alzheimer's Disease and Associated Risk Factors Among Aging Hispanics: A Cross-country Study.”* Accepted for poster presentation at the Alzheimer's Association International Conference 2024, Philadelphia, PA, USA.
- 2024 *“Non-APOE Polygenic risk score derived from European ancestry data are more predictive of dementia among Hispanics who are APOE ϵ 4 carriers.”* Accepted for oral presentation at the Alzheimer's Association International Conference 2024, Philadelphia, PA, USA.
- 2023 *“Childhood self-reported socioeconomic status modifies the genetic risk of Alzheimer's on mid- and late-life cognitive decline in a population-based study.”* Accepted for poster presentation at the Alzheimer's Association International Conference 2023, Amsterdam, Netherlands.
- 2023 *“Childhood self-reported socioeconomic status modifies the genetic risk of Alzheimer's on mid- and late-life cognitive decline in a population-based study.”* Accepted for poster presentation at the Society for Epidemiology Annual Conference 2023, Portland, OR, USA.
- 2023 *“Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals from the Wisconsin Registry for Alzheimer's Prevention.”* Accepted for poster presentation at the Wisconsin Alzheimer's Disease Research Day 2023, Madison, WI, USA.
- 2022 *“Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals from the Wisconsin Registry for Alzheimer's Prevention.”* Accepted for oral presentation at the Synergies for Longitudinal Study Conference, Madison, WI, USA.
- 2022 *“Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals from the Wisconsin Registry for Alzheimer's Prevention.”* Accepted for oral presentation at the Alzheimer's

Association International Conference 2022, San Diego, CA, USA.

- 2022 *“Apolipoprotein E moderates the association between Non-APOE Polygenic Risk Score for Alzheimer’s Disease and Aging on Preclinical Cognitive Function.”*
Accepted for poster presentation at the Alzheimer’s Association International Conference 2022, San Diego, CA, USA.
- 2022 *“Apolipoprotein E moderates the association between Non-APOE Polygenic Risk Score for Alzheimer’s Disease and Aging on Preclinical Cognitive Function.”*
Accepted for poster presentation at the Wisconsin Alzheimer’s Disease Research Day 2022, Madison, WI, USA.
- 2021 *“Effect of Pathway-specific Polygenic Risk Scores for Alzheimer’s Disease on Rate of Change in Cognitive Function and AD-related Biomarkers among Asymptomatic Individuals.”* Accepted for oral presentation at the Integrating Genetics into the Social Sciences Conference 2021, Boulder, CO, USA.
- 2020 *“Bald is Beautiful? Life course outcomes for individuals at high genetic risk for male pattern baldness.”* Accepted for oral presentation at the Integrating Genetics into the Social Sciences Conference 2020, Boulder, CO, USA.

EXTERNAL TRAINING, CERTIFICATES, AND WORKSHOPS

- 2022 **Novel models for longitudinal continuous outcomes workshop**
Stanford University
- 2021 **DNA methylation analysis workshop**
University of Colorado-Boulder
- 2020 **Statistical workshop on genomic family data**
University of Colorado-Boulder
- 2020 **Genome-wide data analyses**
Tinbergen Institute and
University of Wisconsin-Madison

GRANT SUPPORT

- 2022-2023 **Co-Investigator** (PI: Corinne Engelman)
Gene-Environment Interactions in Cognitive Decline During Preclinical Alzheimer’s, UW-Madison CDHA pilot project award, P30AG017266
Total costs: \$50,000

This pilot project aims to understand how early-to-late life environments modify the genetic risk of Alzheimer’s disease (AD) on later-life cognition among asymptomatic individuals. This award directly supports my dissertation research, and my responsibilities include drafting grant proposals, performing relevant

analyses, and drafting the final manuscript. The results from this project will serve as preliminary data for a future R01 grant application.

HONORS/AWARDS

2021, 2023	Graduate Student Research Competition Grant University of Wisconsin-Madison
2022	Rankin / Skatrud Research Award University of Wisconsin-Madison
2019	Chancellor's Fellowship University of Wisconsin-Madison
2019	PHS Outstanding Student Scholarship University of Wisconsin-Madison
2018	Phi Alpha Alpha National Honor Society (inducted in 2018) Network of Schools of Public Policy, Affairs, and Administration
2015	Outstanding Undergraduate Thesis Award Shandong University
2013	Scholarship for Academic Excellence Shandong University

PROFESSIONAL EXPERIENCE

2017 - 2018	Part-time policy analyst (Master's Practicum / Capstone Project) International Refugee Assistance Project
Summer 2017	Policy analyst intern Fudan University PPIR Center

SELECTED MEDIA COVERAGE

2025	TaubCONNECT Research Perspectives: October 2025 https://columbianeuroresearch.org/taub/res-taub-connect-10-2025.html#4
2025	TaubCONNECT Research Perspectives: December 2025 https://columbianeuroresearch.org/taub/res-taub-connect.html

REFEREE (PEER REVIEW)

Scientific Report

Journal of Alzheimer's Disease
Journal of Alzheimer's Disease Report
Brain, Behavior, and Immunity – Health
International Journal of Molecular Sciences
Dementia and Geriatric Cognitive Disorders

PROFESSIONAL MEMBERSHIPS

International Society to Advance Alzheimer's Research and Treatment
Society for Epidemiologic Studies
International Genetic Epidemiology Society

ADDITIONAL INFORMATION

Nationality: People's Republic of China
ORCID: 0000-0002-2732-7199
Software: R, SAS, STATA, Arc-GIS, PLINK, GCTA, LDSC
Programming languages: R (proficient), Python (proficient)
Computational environment: High-performance computing (HPC) on Linux-based clusters (Slurm) for large-scale genomic and biomarker analyses
NCBI Bibliography: <https://www.ncbi.nlm.nih.gov/myncbi/yuexuan.xu.3/bibliography/public/>

REFERENCES

Richard Mayeux, MS, MD

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Co-director, Taub Institute for Research on Alzheimer's Disease and the Aging Brain
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